Does a VWAP + RSI Divergence Strategy Beat QQQ?

October 16, 2026 · Muhsin Bas

Abstract

I set out to answer one question: could this trading strategy make more money than simply buying QQQ and holding it? The strategy mixes two TradingView tools — a ‘fair price’ line (VWAP) and a momentum warning signal (RSI divergence). I tested 1,000 versions of it, at three speeds (daily over 10 years, hourly over 2 years, and 5-minute over 2 months), and ran strict checks to see whether any winner was real. The answer was no. Only about 1 in 100 versions beat buy-and-hold, half of them lost money, and the best one failed every test designed to catch false winners — it looked good only because I tried so many versions. A faster version built on two other indicators lost money even with zero trading fees. Bottom line: the strategy has no real, dependable edge over just holding QQQ.

Background and strategy

The main strategy mixes two indicators from TradingView. Here is what each one does, in plain terms:

- **VWAP — a ‘fair price’ line for the day.** It is the average price so far, weighted by how much was traded. When price is above this line, buyers are in control (bullish); below it, sellers are (bearish). It restarts each week, month, or quarter.
- **RSI divergence — a momentum ‘running out of gas’ warning.** RSI measures whether recent moves are strong or fading. A bullish warning is when price drops to a new low but RSI does not — a hint the fall is losing steam and may turn up. A bearish warning is the mirror image at a new high.

The trading rule combined them as a trigger-plus-filter: an RSI divergence triggers a trade and the VWAP regime filters its direction. Four combination modes were tested (divergence-only, divergence + VWAP filter, VWAP-only, VWAP + divergence-confirm), each long-only / short-only / both, with two exit styles.

A second, higher-frequency strategy paired the **Naive Bayes classifier** from the 'Monte Carlo CT' indicator (features: relative volume and 10-bar momentum) with **John Ehlers' Adaptive SuperSmoother** (TASC 2026.09) as a low-lag trend filter.

Data and methodology

Data. QQQ price history from Yahoo Finance (yfinance), dividend-adjusted. Daily bars reach 10 years; intraday history is capped by the provider — hourly to ~2 years, 5/15/30-minute to ~60 days. Ten years of intraday data is not available for free, so the daily test is the only true decade-long backtest.

Search. Each run sampled 1,000 strategy configurations at random across the knob space: combination mode, direction, exit style, VWAP anchor, RSI length (7–21), pivot lookback (3–8), and hidden divergences on/off.

No lookahead. A divergence is acted on only at the bar where its pivot confirms (a built-in lag equal to the pivot lookback), and every position earns the *next* bar's return, never the current bar's.

Costs. Trading cost is modeled as basis points per position change and swept from 0 to 10 bps to test sensitivity.

Buy-and-hold of QQQ over the same window is the benchmark in every test.

Results: 10-year daily

Buy-and-hold QQQ returned **+567%** over the decade (2016–2026). Of 1,000 strategies, only **9 (0.9%)** beat it, the **median strategy lost 2.4%**, and the worst lost 91%.

The single best configuration returned **+702%** — but it was vwap_only / long_only, meaning it ignored the RSI divergence entirely and stayed long ~96% of the time. The best 'strategy' was buy-and-hold in disguise, lightly sidestepping a few dips.

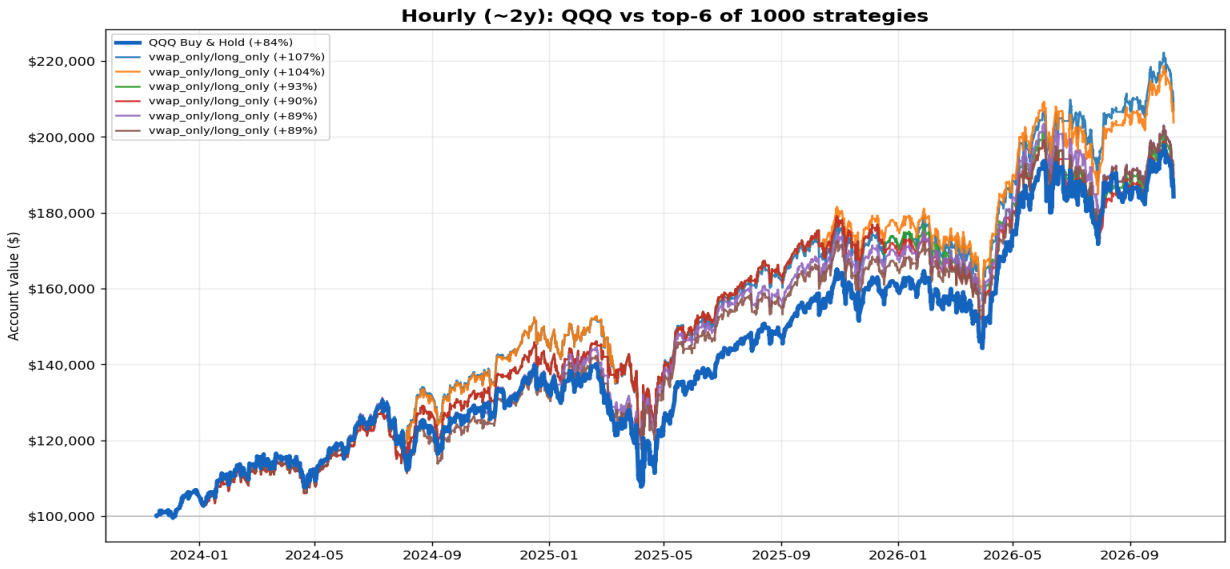


Plotted against all 1,000, the vast majority fall well below the QQQ benchmark; only a thin top edge keeps pace, and every strategy on that edge is a near-permanent-long configuration.



Results: hourly (~2 years)

On ~2 years of hourly bars, QQQ returned **+84%**. Again only **9 of 1,000** strategies beat it, and the median lost money. The best hourly configuration (**+107%**) was once more vwap_only / long_only — a closet-long position, not a divergence strategy.



Results: 5-minute high-frequency attempt

I also tested a faster version on 5-minute bars (about 60 days of data — the most the data provider offers at that speed). It paired a 'Naive Bayes' guesser with Ehlers' Adaptive SuperSmoother trend line. It failed badly: **even with no trading fees at all, it lost 17%**, while QQQ gained 3% over the same stretch. Once realistic fees were added it lost nearly half its money (**–49%**), because it traded far too often — **907 times in 60 days**. At that speed the signals are basically noise, and paying a fee on every one of hundreds of trades locks in losses.

The signal had no edge at this frequency: momentum and relative volume on 5-minute bars behave as noise, and every trade pays the spread. True microsecond HFT cannot be expressed in Pine Script or paper-traded in MetaTrader, so this was always 'fast intraday', not HFT.

Robustness testing

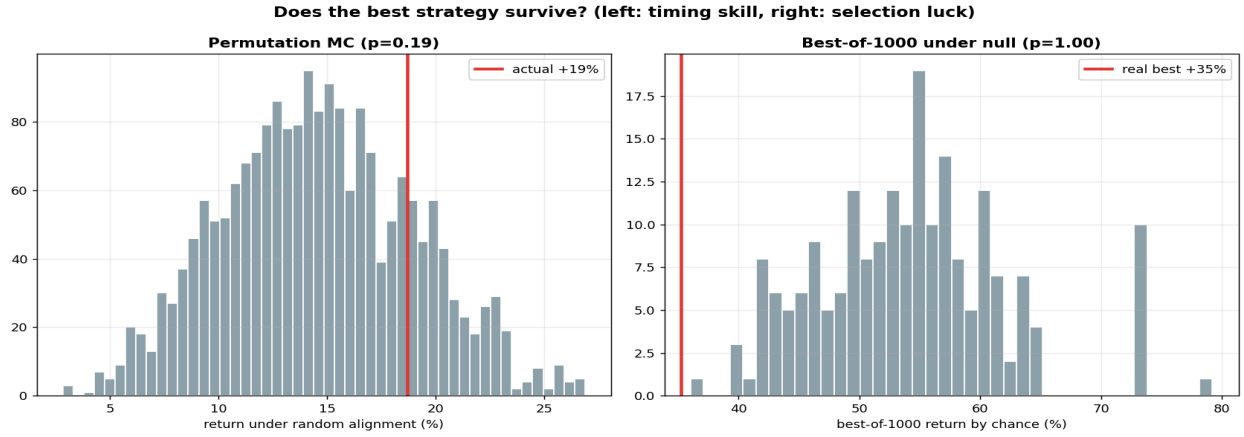
Three tests decide whether an apparent winner is real. The best hourly configuration was put through all three.

Out-of-sample. Run on windows it was *not* selected on, the config was inconsistent — it beat QQQ in some periods and lost in others, with no reliability:

Window	Strategy	QQQ	Result
Pre-Trump 2023–24	+47.5%	+35.4%	beat
Trump 1st year 2025	+17.1%	+18.8%	lost
Trump 2nd year (selected)	+18.8%	+15.9%	beat

Permutation Monte Carlo. Shuffling the return series relative to the positions 2,000 times gives a p-value of **0.19** — the timing is not statistically distinguishable from random alignment.

The 'too many tries' test (the decisive one). When you try 1,000 strategies, some will look great by pure luck — like flipping 1,000 coins and finding one that lands on heads 10 times in a row. To check for this, I scrambled all 1,000 strategies so none of them could have any real skill, then took the best of the bunch. By luck alone, that best scrambled strategy averaged **+54%** — actually *better* than my real best of +35%. So trying 1,000 versions fully explains my result: the 'winner' wasn't skill, it was simply the luckiest of many guesses.



Discussion and limitations

The same thing kept happening across every test: the only strategies that beat QQQ were ones that stayed long almost the entire time. They didn't win by timing trades well — they won simply by holding a market that kept rising. The strategies that actually relied on the RSI divergence signals mostly did worse than just buying and holding. One version looked like a real winner in the choppy 2026 market, but once I accounted for the fact that I had tried 1,000 different versions, even that one turned out to be luck, not skill.

Limitations. Intraday history is shallow (~2 years hourly, ~60 days for minute bars), so intraday samples are small. Backtests assume fills at closing prices with simple basis-point costs; real slippage, bid-ask spread, and CFD overnight financing are not fully modeled. Results are specific to QQQ over 2016–2026, a period dominated by a strong bull market, which structurally favors buy-and-hold. Free Yahoo Finance data may contain minor adjustments or gaps.

Conclusion

No robust, tradeable edge was found. Across 1,000 configurations and multiple time frames, the strategy did not reliably beat buying and holding QQQ; the apparent winners were selection luck and closet indexing, and the high-frequency variant lost money before costs.

This is a successful backtest, not a failed one — its purpose was to learn whether the edge is real *before* risking money, and it answered clearly. The only honest path to validating a future variant is **walk-forward testing**: select a configuration on a training window, test it on the next untouched window, roll forward, and trust only a configuration that wins across several out-of-sample periods. Until a strategy passes that bar, paper-trade it with no expectation of matching the backtest, and do not commit real capital.

References

1. Fabio. *AMT VWAP [Fabio]*. TradingView Pine Script indicator (community script). Retrieved from https://www.tradingview.com
2. TradingView. *RSI Divergence Indicator*. Built-in Pine Script indicator. Retrieved from https://www.tradingview.com
3. Steversteves. *Monte Carlo CT [SS]*. TradingView Pine Script indicator (community script). Retrieved from https://www.tradingview.com
4. Ehlers, J. F. (2026, September). Improved filter performance: Adaptive SuperSmoother. *Technical Analysis of Stocks & Commodities.* Pine Script implementation *TASC 2026.09 Adaptive SuperSmoother* published by PineCoders on TradingView, https://www.tradingview.com
5. yfinance (Python library). Yahoo! Finance market data downloader. https://github.com/ranaroussi/yfinance
6. Yahoo Finance. QQQ (Invesco QQQ Trust) historical price data, 2016–2026. https://finance.yahoo.com
7. Python data/analysis libraries: pandas, NumPy, and Matplotlib.